

James Nithil V

+91 8220173595 | jamnithil@gmail.com | jamesnithil.vercel.app | linkedin/jamesnithil-v | github/nithiljn | leetcode/NITHIL07

EXPERIENCE

Junior Software Developer

Aug 2025 to Present

Vaken Technology | Product: Sovablu

Trichy, India

- Engineered and scaled enterprise RESTful APIs using **Java 21, Spring Boot, and PostgreSQL**, implementing strict **Spring Transaction management and ACID principles** across critical mutation and confirmation workflows to maintain data consistency and prevent data loss under concurrent traffic.
- Architected a hybrid persistence layer pairing **PostgreSQL** with **Amazon DynamoDB** for high-speed key-value reads; restructured table designs and eliminated redundant columns to reduce I/O payload size while implementing **composite indexing and query optimization** to eliminate full row scans.
- Implemented an in-memory **Redis caching tier with TTL-based expiration** for hot-path datasets to reduce repeated database queries; applied **cache-stampede prevention mechanisms** using mutex locks to safeguard the database from concurrent traffic spikes and improve API response times.
- Secured file and document storage using **Amazon S3 with time-limited pre-signed URLs** and enforced **Role-Based Access Control (RBAC)** across platform endpoints; automated Java compilation, testing, and containerized deployment to **Amazon ECS** via **Jenkins CI/CD** with **AWS Lambda** for event-driven scheduled tasks.
- Contributed to **Agentic AI development** by building autonomous agents, defining modular skills, integrating **MCP-based tool access**, and deploying **RAG pipelines with guardrails**; optimized LLM inference costs and response efficiency through **task-based model routing**, directing routine operations to lightweight models and reserving complex reasoning for advanced LLMs.

PROJECTS

KadalVazhi: Autonomous Maritime Intelligence Platform

2026

Java, Spring Boot, Python, LangGraph, Kafka, PostgreSQL, DynamoDB, gRPC

Production Microservices

- Architected distributed maritime engine with **Java** and **Spring Boot** for core backend security and **Python** for **LangGraph multi agent orchestration** and **Claude SDK model routing**.
- Engineered microservices communication layer utilizing **Spring Boot, REST APIs, GraphQL, and gRPC** for high throughput vessel telemetry, direct fish trade, and automated **emergency mechanic dispatch agents** for at sea boat repairs.
- Implemented **dual database architecture with PostgreSQL and DynamoDB**, backed by **Apache Kafka** event streaming to scale concurrent vessel telemetry and power dynamic **fisherman and boat crew job matchmaking**.
- Designed resilient **offline first synchronization engine**: logs vessel telemetry in local SQLite at sea and auto syncs to cloud PostgreSQL upon cellular reconnection, with a 3 attempt fallback loop for weather safety.
- Built comprehensive fish catch analytics engine aggregating historical catch data for data driven voyage decisions, supporting multilingual accessibility across **Tamil, English, and Malayalam via i18n**.

TECHNICAL SKILLS

Languages: Java (Core & 21), Python, SQL, C++, JavaScript

Backend & APIs: Spring Boot, FastAPI, REST APIs, GraphQL, gRPC, Microservices

Distributed Systems & Cloud: Apache Kafka, Distributed Systems, Redis Caching, Rate Limiting, Docker, Jenkins CI/CD, AWS (EC2, S3, Lambda, Bedrock)

Databases: PostgreSQL, DynamoDB, MySQL, Redis, SQLite

AI & Systems: LangGraph, Agentic AI, Multi Agent Orchestration, Claude SDK, LangSmith, i18n

ACHIEVEMENTS

LeetCode Knight Badge

Global Rank #114 / 43,000+

Contest Rating 1,994 (Top 2.73% Global)

LeetCode

- Earned the **LeetCode Knight badge** (Top 2.73% worldwide) with an official **Contest Rating of 1,994**; achieved multiple 4/4 perfect solves with a peak **Global Rank #114** in Weekly Contest 517 and **Rank #197** in Contest 490.
- Solved **700+ algorithmic problems**: 446 in **Java** (Graphs, Trees, DP, BFS/DFS, Linked Lists), 443 in **Python3**, 40 in **MySQL/SQL**, and 31 in **C++**.

EDUCATION

Knowledge Institute of Technology

Salem, India

B.Tech in Artificial Intelligence and Data Science | CGPA: 8.1

2021 to 2025

CERTIFICATIONS

Core Java & SQL (EBox) | MySQL Database (IBM) | Machine Learning (Simplilearn) | Deep Learning PyTorch (Guvi)